

SE3040 - Application Frameworks

SE-132

Introduction

AquaShield is a full-stack web application developed as a group academic project. Its primary mission is to help protect marine life by monitoring illegal fishing activities, managing information on protected marine species, and streamlining the process of handling reported incidents.

Key Features

- **User & Admin Roles:** The system supports both regular users (who can submit reports) and administrators (who can manage users, cases, and species data).
- **Incident Reporting & Case Management:** Users can submit reports of illegal activities along with evidence (like images). Administrators can then review these reports, manage their statuses, and handle official "cases" based on those reports.
- **Marine Species Tracking:** The application maintains a database of protected species. It even integrates with a third-party biological database (GBIF) to enrich its species data with extra details and media.
- **Authentication & Security:** Uses JSON Web Tokens (JWT) and sessions for secure login, registration, and role-based access control.
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Technology Stack

- **Frontend:** Built with React (using Vite for fast builds), functional routing via React Router, and styled beautifully with Tailwind CSS.
- **Backend:** Powered by Node.js and Express.js to create a robust REST API.
- **Database:** MongoDB (using Mongoose) for flexible, document-based data storage.
- **Additional Tools:** Uses Multer for file uploads, Nodemailer for emails, and external APIs for geocoding and species data.
- **Deployment:** Designed to be deployed with the backend on Render and the frontend on Vercel.

1. Report Management API Documentation

Base Path: /api/reports

- **Create Report**

Authentication: Required (User)

Method: POST

URL:

http://localhost:5000/api/reports

Body (form-data)

```
{
  "incidentType": "Illegal Net Fishing",
  "description": "Fishing activity observed in protected marine zone...",
  "latitude": 6.9271,
  "longitude": 79.8612,
  "incidentDate": "2026-04-10",
  "speciesInvolved": ["speciesId1", "speciesId2"]
}
```

Files (optional)

evidence: [image1.jpg, video1.mp4]

Expected Status

201 Created

Expected Response

```
{
  "message": "Report submitted successfully.",
  "report": {
    "_id": "69xxxxxx",
    "reportedBy": "userId",
    "incidentType": "Illegal Net Fishing",
    "description": "Fishing activity observed...",
    "location": {
      "type": "Point",
      "coordinates": [79.8612, 6.9271]
    },
    "incidentDate": "2026-04-10T00:00:00.000Z",
    "evidence": [
```

```
{
  "url": "/uploads/file.jpg",
  "fileType": "image/jpeg",
  "originalName": "file.jpg"
},
"speciesInvolved": [],
"status": "Pending",
"adminNote": "",
"createdAt": "2026-04-10T10:00:00.000Z",
"updatedAt": "2026-04-10T10:00:00.000Z"
}
```

• Get My Reports

Authentication: Required (User)

Method: GET

<http://localhost:5000/api/reports/my>

Expected Status

200 OK

Expected Response

```
{
  "count": 2,
  "reports": [
    {
      "_id": "69xxxxxx",
      "incidentType": "Illegal Net Fishing",
      "description": "...",
      "status": "Pending"
    }
  ]
}
```

• Get My Report By ID

Authentication: Required

Method: GET

<http://localhost:5000/api/reports/my/{id}>

Expected Status

200 OK

Expected Response

```
{
  "_id": "69xxxxxx",
  "incidentType": "Illegal Net Fishing",
  "status": "Pending"
}
```

• Update My Report

Authentication: Required

Method: PUT

<http://localhost:5000/api/reports/my/{id}>

Rules

- Only **Pending** reports can be updated

Body (raw JSON or form-data)

```
{
  "description": "Updated description",
  "latitude": 6.93,
  "longitude": 79.88,
  "incidentType": "Night Fishing Violation"
}
```

Expected Status

200 OK

Expected Response

```
{
  "message": "Report updated successfully.",
  "report": { }
}
```

• Delete My Report

Authentication: Required

Method: DELETE

http://localhost:5000/api/reports/my/{id}

Rules

- Only **Pending** reports can be deleted

Expected Status

200 OK

Response

```
{
  "message": "Report deleted successfully."
}
```

• Admin - Get All Reports

Authentication: Admin Required

Method: GET

http://localhost:5000/api/reports

Query Params (optional)

status=Pending
page=1
limit=10

Expected Response

```
{
  "total": 10,
  "page": 1,
  "pages": 2,
  "reports": []
}
```

• Admin - Get Report By ID

Authentication: Admin Required

Method: GET

http://localhost:5000/api/reports/{id}

Expected Status

200 OK

- **Admin - Update Report Status**

Authentication: Admin Required

Method: PATCH

`http://localhost:5000/api/reports/{id}/status`

Body

```
{  
  "status": "Verified",  
  "adminNote": "Verified after investigation"  
}
```

Valid Status Values

Pending
Under Review
Verified
Dismissed
Resolved

Expected Response

```
{  
  "message": "Report status updated.",  
  "report": {}  
}
```

2. User Management API Documentation

Base Path: `/api/users`

- **Create User**

Authentication: Not required (Public)

Method: POST

`http://localhost:5000/api/users`

Body (raw JSON)

```
{
  "firstName": "Kasun",
  "lastName": "Perera",
  "email": "kasun.perera@gmail.com",
  "password": "123456"
}
```

Expected Status

200 OK

Expected Response

```
{
  "message": "User created successfully",
  "token": "JWT_TOKEN",
  "user": {
    "id": "USER_ID",
    "email": "kasun.perera@gmail.com",
    "firstName": "Kasun",
    "lastName": "Perera",
    "isAdmin": false,
    "isBlocked": false,
    "image": "/images/default-profile.png"
  }
}
```

Status: Pass

Create User – Missing Fields

Method: POST

http://localhost:5000/api/users

Body

```
{
  "firstName": "Kasun"
}
```

Expected Status

400 Bad Request

Expected Response

```
{
  "errors": [
    { "msg": "Email is required" },
    { "msg": "Last name is required" },
    { "msg": "Password is required" }
  ]
}
```

Status: Pass

• Create User – Duplicate Email

Method: POST

http://localhost:5000/api/users

Body

```
{
  "firstName": "Nimali",
  "lastName": "Fernando",
  "email": "kasun.perera@gmail.com",
  "password": "123456"
}
```

Expected Status

400 Bad Request

Expected Response

```
{
  "message": "Email already exists"
}
```

Status: Pass

• Login User

Method: POST

http://localhost:5000/api/users/login

Body


```
{
  "email": "kasun.perera@gmail.com",
  "password": "123456"
}
```

Expected Status

200 OK

Expected Response

```
{
  "message": "Login successful",
  "token": "JWT_TOKEN"
}
```

Status: Pass

• Login – Wrong Password

Method: POST

http://localhost:5000/api/users/login

Body

```
{
  "email": "kasun.perera@gmail.com",
  "password": "wrong123"
}
```

Expected Status

401 Unauthorized

Expected Response

```
{
  "message": "Invalid password"
}
```

Status: Pass

• Logout User

Authentication: Required

Method: POST

http://localhost:5000/api/users/logout

Expected Status

200 OK

Response

```
{  
  "message": "Logged out successfully"  
}
```

Status: Pass

• Get All Users (Admin)

Authentication: Admin Required

Method: GET

http://localhost:5000/api/users

Expected Status

200 OK

Response

```
[  
  {  
    "_id": "USER_ID",  
    "email": "kasun.perera@gmail.com",  
    "firstName": "Kasun",  
    "lastName": "Perera",  
    "isAdmin": false,  
    "isBlocked": false  
  }  
]
```

Status: Pass

• Search User (Admin)

Method: GET

`http://localhost:5000/api/users/search?query=kasun`

Expected Status

200 OK

Response

```
[
  {
    "email": "kasun.perera@gmail.com",
    "firstName": "Kasun",
    "lastName": "Perera"
  }
]
```

Status: Pass

• Update Profile

Authentication: Required

Method: PUT

`http://localhost:5000/api/users/me`

Body

```
{
  "firstName": "Kasun",
  "lastName": "Silva"
}
```

Expected Status

200 OK

Response

```
{
  "message": "Profile updated successfully",
  "user": {
    "firstName": "Kasun",
    "lastName": "Silva"
  },
  "token": "NEW_TOKEN"
}
```

Status: Pass

• Update Profile – Restricted Fields

Method: PUT

http://localhost:5000/api/users/me

Body

```
{
  "isAdmin": true
}
```

Expected Status

403 Forbidden

Response

```
{
  "message": "You cannot change system fields"
}
```

Status: Pass

• Delete Own Account

Authentication: Required

Method: DELETE

http://localhost:5000/api/users/me

Expected Status

200 OK

Response

```
{  
  "message": "Account deleted successfully"  
}
```

Status: Pass

• Forgot Password (OTP)

Method: POST

http://localhost:5000/api/users/forgot-password

Body

```
{  
  "email": "kasun.perera@gmail.com"  
}
```

Expected Status

200 OK

Response

```
{  
  "message": "OTP sent to your email"  
}
```

Status: Pass

• Reset Password

Method: POST

http://localhost:5000/api/users/reset-password

Body

```
{  
  "email": "kasun.perera@gmail.com",  
  "otp": "123456",  
  "newPassword": "newpassword123"  
}
```

Expected Status

200 OK

Response

```
{  
  "message": "Password reset successfully"  
}
```

Status: Pass

• Block User (Admin)

Method: PUT

http://localhost:5000/api/users/block/USR-0001

Expected Response

```
{  
  "message": "User blocked successfully"  
}
```

Status: Pass

Unblock User (Admin)

Method: PUT

http://localhost:5000/api/users/unblock/USR-0001

Expected Response

```
{  
  "message": "User unblocked successfully"  
}
```

Status: Pass

3. Case Management API Test Documentation

Base Path: /api/cases

- **Create Case (Admin Only)**

Authorization: Bearer <token> (Admin only)

Method: POST

http://localhost:5000/api/cases

Body (raw JSON)

```
{
  "caseNumber": "CASE-001",
  "reportId": "REPLACE_WITH_REPORT_ID",
  "assignedOfficer": "Wildlife Officer Colombo",
  "status": "OPEN",
  "priority": "HIGH",
  "notes": "Illegal fishing activity reported in Kelani River area"
}
```

Expected Status

201 Created

Expected Response

```
{
  "_id": "case_id_here",
  "caseNumber": "CASE-001",
  "reportId": "report_id_here",
  "locationName": "Kelani River, Sri Lanka",
  "status": "OPEN",
  "priority": "HIGH",
  "assignedOfficer": "Wildlife Officer Colombo",
  "notes": "Illegal fishing activity reported in Kelani River area",
  "createdAt": "2026-04-11T00:00:00.000Z",
  "updatedAt": "2026-04-11T00:00:00.000Z"
}
```

Status: Pass

● Create Another Case (High Priority Wildlife Violation)

Authorization: Bearer <token> (Admin only)

Method: POST

http://localhost:5000/api/cases

Body

```
{
  "caseNumber": "CASE-002",
  "reportId": "REPLACE_WITH_REPORT_ID",
  "assignedOfficer": "Marine Conservation Unit - Galle",
  "status": "UNDER_INVESTIGATION",
  "priority": "HIGH",
  "notes": "Illegal coral fishing detected near Galle coastal area"
}
```

Expected Status

201 Created

Expected Response

```
{
  "message": "Case created successfully",
  "case": {
    "_id": "case_id_here",
    "caseNumber": "CASE-002",
    "status": "UNDER_INVESTIGATION",
    "priority": "HIGH"
  }
}
```

Status: Pass

● Get All Cases (Admin Only)

Authorization: Bearer <token> (Admin only)

Method: GET

http://localhost:5000/api/cases

Expected Status

200 OK

Expected Response

```
[
  {
    "_id": "case_id_1",
    "caseNumber": "CASE-001",
    "locationName": "Kelani River, Sri Lanka",
    "status": "OPEN",
    "priority": "HIGH"
  },
  {
    "_id": "case_id_2",
    "caseNumber": "CASE-002",
    "locationName": "Galle Coastal Area, Sri Lanka",
    "status": "UNDER_INVESTIGATION",
    "priority": "HIGH"
  }
]
```

Status: Pass

• Get Case By ID

Authorization: Bearer <token>

Method: GET

http://localhost:5000/api/cases/:id

Example

http://localhost:5000/api/cases/CASE_ID_HERE

Expected Status

200 OK

Expected Response

```
{
  "_id": "case_id_here",
  "caseNumber": "CASE-001",
  "reportId": {
    "_id": "report_id",
    "incidentType": "Illegal Net Fishing",
    "location": {
      "coordinates": [79.8612, 6.9271]
    }
  }
},
```

```
"locationName": "Kelani River, Sri Lanka",  
"status": "OPEN",  
"priority": "HIGH"  
}
```

Status: Pass

• Update Case (Admin Only)

Authorization: Bearer <token> (Admin only)

Method: PUT

http://localhost:5000/api/cases/:id

Body

```
{  
  "status": "LEGAL_ACTION_STARTED",  
  "priority": "MEDIUM",  
  "assignedOfficer": "Legal Enforcement Unit - Colombo"  
}
```

Expected Status

200 OK

Expected Response

```
{  
  "message": "Case updated successfully",  
  "case": {  
    "_id": "case_id_here",  
    "status": "LEGAL_ACTION_STARTED",  
    "priority": "MEDIUM"  
  }  
}
```

Status: Pass

• Delete Case (Admin Only)

Authorization: Bearer <token> (Admin only)

Method: DELETE

http://localhost:5000/api/cases/:id

Expected Status

200 OK

Expected Response

```
{  
  "message": "Case deleted successfully"  
}
```

Status: Pass

- Case creation: Admin only
- Case tracking: Linked with Reports
- Status workflow supported:
 - OPEN
 - UNDER_INVESTIGATION
 - LEGAL_ACTION_STARTED (etc.)
- Full CRUD operations supported
- Strong enforcement for wildlife violation tracking system

4. Species Management API Test Documentation

Base Path: /api/species

• Create Species

Authentication: Bearer <token> (Admin only)

Method: POST

http://localhost:5000/api/species

Body (raw JSON)

```
{  
  "id": "SL-FISH-001",
```

```
"name": "Sri Lankan Mahseer",
"scientificName": "Tor tor",
"bodyShape": "fusiform",
"tailShape": "forked",
"finType": "continuous",
"colorPattern": "golden-brown with dark fins",
"protectionStatus": "Endangered",
"isFullyBanned": true,
"legalMinSizeCm": 0,
"legalSeason": {
  "startMonth": 1,
  "endMonth": 12
},
"regions": [
  "Kelani River",
  "Mahaweli River",
  "Gin River"
],
"image": "https://example.com/sri-lankan-mahseer.jpg",
"description": "Endemic freshwater fish species found in Sri Lankan rivers"
}
```

Expected Status

201 Created

Expected Response

```
{
  "message": "Species Created Successfully"
}
```

Status: Pass

• Create Another Species

Method: POST

http://localhost:5000/api/species

Body

```
{
  "id": "SL-FISH-002",
  "name": "Green Chromide",
  "scientificName": "Etroplus suratensis",
  "bodyShape": "oval",
  "tailShape": "rounded",
  "finType": "continuous",
}
```

```
"colorPattern": "greenish-yellow with spots",
"protectionStatus": "Legal",
"isFullyBanned": false,
"legalMinSizeCm": 15,
"legalSeason": {
  "startMonth": 5,
  "endMonth": 9
},
"regions": [
  "Negombo Lagoon",
  "Chilaw Lagoon",
  "Jaffna Lagoon"
],
"image": "https://example.com/green-chromide.jpg",
"description": "Brackish water fish common in Sri Lankan lagoons"
}
```

Status: Pass

• Get All Species

Method: GET

<http://localhost:5000/api/species>

Expected Status

200 OK

Response

```
[
  {
    "_id": "69a65e41bb06087b808766d7",
    "id": "SL-FISH-001",
    "name": "Sri Lankan Mahseer",
    "scientificName": "Tor tor",
    "bodyShape": "fusiform",
    "tailShape": "forked",
    "finType": "continuous",
    "colorPattern": "golden-brown with dark fins",
    "protectionStatus": "Endangered",
    "isFullyBanned": true,
    "legalMinSizeCm": 0,
    "legalSeason": {
      "startMonth": 1,
      "endMonth": 12
    },
    "regions": [
      "Kelani River",

```

```
"Mahaweli River",  
"Gin River"  
],  
"image": "https://example.com/sri-lankan-mahseer.jpg",  
"description": "Endemic freshwater fish species found in Sri Lankan rivers"  
}  
]
```

Status: Pass

• Get Species By ID

Method: GET

http://localhost:5000/api/species/SL-FISH-001

Status: Pass

• Update Species

Authentication: Admin only

Method: PUT

http://localhost:5000/api/species/SL-FISH-001

Body

```
{  
  "name": "Sri Lankan Golden Mahseer",  
  "legalMinSizeCm": 30,  
  "isFullyBanned": false  
}
```

Expected Response

```
{  
  "message": "Species Updated Successfully",  
  "species": {  
    "_id": "69a65e41bb06087b808766d7",  
    "id": "SL-FISH-001",  
    "name": "Sri Lankan Golden Mahseer",  
    "scientificName": "Tor tor",  
    "bodyShape": "fusiform",  
    "tailShape": "forked",  
    "finType": "continuous",
```

```
"colorPattern": "golden-brown with dark fins",
"protectionStatus": "Endangered",
"isFullyBanned": false,
"legalMinSizeCm": 30
}
}
```

Status: Pass

• Delete Species

Authentication: Admin only

Method: DELETE

<http://localhost:5000/api/species/SL-FISH-004>

Response

```
{
  "message": "Species Deleted Successfully"
}
```

Status: Pass

• Filter Species (Morphology Search)

Method: GET

<http://localhost:5000/api/species/find>

Query Example

?bodyShape=fusiform&tailShape=forked&finType=continuous&colorPattern=golden

Response

```
{
  "message": "Species found successfully",
  "matchCount": 2,
  "selectedFilters": {
    "bodyShape": "fusiform",
    "tailShape": "forked",
    "finType": "continuous",
    "colorPattern": "golden"
  }
}
```

```
},
"species": [
  {
    "id": "SL-FISH-001",
    "name": "Sri Lankan Golden Mahseer",
    "scientificName": "Tor tor"
  }
]
}
```

Status: Pass

• Filter by Shape

Method: GET

<http://localhost:5000/api/species/find?bodyShape=oval&tailShape=rounded>

Status: Pass

• GBIF Integration – Search Species

Method: GET

<http://localhost:5000/api/species/gbif/search?q=Tor+tor+Sri+Lanka>

Response

```
{
  "count": 8,
  "results": [
    {
      "gbifKey": 2375999,
      "scientificName": "Tor tor",
      "canonicalName": "Tor tor",
      "family": "Cyprinidae",
      "order": "Cypriniformes",
      "class": "Actinopterygii",
      "kingdom": "Animalia"
    }
  ]
}
```

Status: Pass

- **GBIF Enriched Data**

Method: GET

<http://localhost:5000/api/species/gbif/2375999>

Status: Pass

- **Local + GBIF Enriched Species**

Method: GET

<http://localhost:5000/api/species/SL-FISH-001/enrich>

Response Includes:

- Local species data
- GBIF taxonomy data
- Images & occurrence data

Status: Pass